

MEHUL UNIYAL

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PROFESSIONAL SUMMARY

AI/ML Engineer with production experience across generative AI, computer vision, and retrieval-augmented generation. Currently building Azure-based RAG chatbots with streamlined data ingestion, large-scale deep web-crawling pipelines, and a computer-vision-based compliance-monitoring system that analyzes CCTV and 360° camera feeds using vision-language models (VLM) and YOLO object detection. Previously sole backend architect of a multi-tenant voice AI platform routing live calls across 6+ AI bots with a pluggable 5-TTS and 3-STT adapter factory, achieving sub-200ms TTS latency. Deep expertise in hybrid retrieval (FAISS, BM25, RRF), cross-encoder reranking, LangGraph agents, FSM dialogue management, and end-to-end model training and deployment.

WORK EXPERIENCE

AI/ML Engineer | **Preview Technologies** | Noida, India

Jun 2026 – Present

- Building production RAG chatbots on Microsoft Azure with automated data ingestion pipelines, integrating Azure OpenAI to deliver grounded, context-aware conversational responses over client knowledge bases.
- Engineered a deep website-crawling and content-ingestion pipeline for the UP Eco Tourism project, systematically extracting and structuring large volumes of site content to power the chatbot's retrieval knowledge base.
- Architecting an AI-powered Compliance Management System for the Delhi Pollution Control Committee (DPCC) that analyzes on-site CCTV and 360° camera feeds to automatically verify environmental compliance adherence in real time.
- Training and fine-tuning computer-vision models on custom-annotated datasets for compliance detection — currently deploying vision-language models (VLM) and expanding the pipeline to YOLO-based object detection for real-time monitoring.

GenAI Engineer | **Novus Connect** | Noida, India

May 2025 – Jun 2026

- Architected a multi-project WebSocket call dispatcher routing live telephony traffic across 6+ AI voice bots (recruitment, healthcare, IVR, restaurant), reducing zero-code project onboarding time by 80% via YAML-driven configuration and per-call session isolation.
- Built a pluggable adapter factory supporting 5 TTS engines (Google Neural2, Sarvam Bulbul v3, Azure, ElevenLabs) and 3 STT engines (Google Streaming, Deepgram, WebRTC VAD), enabling full provider switching through a single config change and cutting integration time by ~70%.
- Achieved sub-200ms TTS latency by implementing parallel sentence synthesis (`asyncio.create_task`), thread-pool offloading, persistent HTTP keep-alive, and 40ms PCM chunking over a custom reverse-media WebSocket, improving voice responsiveness across all 6+ bots.
- Constructed a SlotsManager with locked-slot architecture, dual-path extraction (regex + Groq LLM), streaming STT callbacks, and dual-mode barge-in via RMS thresholding and partial transcript matching.
- Deployed a production RAG stack with smart chunking, BGE embeddings, FAISS + BM25 hybrid retrieval (RRF), cross-encoder reranking, and a RAGMixin injecting grounded responses into FSM flows without disrupting slot-filling; semantic query triggering under 1ms via cosine similarity.

PROJECTS

ClinicVoice – AI Voice Receptionist for Healthcare Clinics | *Solo Project*

- Delivered an end-to-end real-time voice appointment agent (booking, rescheduling, cancellation, FAQ) using FastAPI, ElevenLabs STT streaming, Groq LLaMA-3.3-70b, and ElevenLabs TTS, achieving a sub-600ms voice round-trip (STT 150ms, LLM first token 180ms, TTS first audio 160ms).
- Built a browser-to-speaker WebSocket audio pipeline with Redis-backed session state supporting low-latency, stateful multi-turn conversations with LLM guardrails scoped to clinic context.
- Engineered a self-service clinic onboarding pipeline (URL scraper to LLM to PostgreSQL profile), enabling zero-manual-config bot deployment per clinic and reducing setup time from hours to minutes.
- Launched a clinic-facing management portal (Vue 3 + Vite) covering appointment management, schedule views, staff actions, and bot settings, backed by a full DB schema with SQLAlchemy and Alembic migrations.

AI-Powered Call QA and Intent Analytics

- Deployed post-call QA scoring across 5 quality dimensions (empathy, compliance, communication, resolution, tone) using Groq LLM with chunk-based transcript processing and SQL Server persistence, processing 100+ call transcripts per batch.
- Constructed an acoustic stress intelligence system (Parselmouth) extracting F0, jitter, shimmer, HNR, and energy variance with a 0–100 distress scoring scale; integrated Pyannote speaker diarization and spaCy NER for zero-shot intent classification across 4 Indian languages.

Speech Emotion Recognition System

- Trained a CNN-based multi-class emotion classifier on labeled audio datasets using librosa-extracted MFCC, chroma, and spectral contrast features on windowed segments, achieving reliable classification across 6 emotion categories.

Production-Grade Telephony AI Voice Assistant Platform

- Launched a WebSocket-based IVR system processing continuous 8kHz LINEAR16 PCM audio with a FlowRunner/ScriptRunner dispatcher for dialogue state, slot filling, and call lifecycle; shipped as a production wheel with session persistence, fallback handling, and graceful termination.
- Integrated Google STT, Google Cloud TTS, reverse-media streaming, and YAML-based project routing, enabling multi-tenant deployment with zero per-project code changes.

Additional Projects

- **AI Recipe Generator:** Cross-platform app generating recipes from user-input ingredients via Meta LLaMA 2 (Streamlit backend, Kotlin frontend).
- **AI-Powered Speaker Diarization System:** Tool segmenting audio into speaker-specific chunks using Hugging Face Pyannote; tested on multi-speaker recordings with 3+ participants.

TECHNICAL SKILLS

Languages: Python, JavaScript, Kotlin

AI and GenAI: Groq LLMs, OpenAI APIs, Azure OpenAI, LangGraph, LangChain, Hugging Face Transformers, spaCy NER, Librosa, Prompt Engineering

Computer Vision and Model Training: Vision-Language Models (VLM), YOLO, Object Detection, Image and Video Annotation, Dataset Curation, Model Training and Fine-Tuning, CCTV and 360° Video Analytics, OpenCV

RAG and Search: FAISS, BM25 (BM25Okapi), Hybrid Retrieval (RRF), Cross-Encoder Reranking, Sentence Transformers (BGE), FastEmbed, Semantic Similarity, Grounded Generation, Multi-Query Expansion, HyDE

Voice AI and Telephony: Sarvam AI v2.5, Google Cloud STT/TTS (Neural2, Streaming), Deepgram, Whisper, Pyannote, Parselmouth, IVR Systems, Telephony Audio Processing (8kHz PCM), Slot Filling, Barge-In, FSM Dialogue Systems

Backend and APIs: FastAPI, Flask, WebSockets, AsyncIO, REST APIs, YAML/JSON configuration

Data Engineering: Web Crawling and Deep Scraping, Data Ingestion Pipelines, ETL, Pandas, NumPy

Data and Storage: SQL Server (pyodbc), PostgreSQL, MySQL, Redis

Cloud and Deployment: Microsoft Azure, AWS EC2, AWS S3, Docker, Railway, Linux, CI/CD Pipelines

Tooling and Observability: Git, GitHub, GitHub Actions, Langfuse, Postman, Logging, Debugging

Frontend: React (v18+), Vue 3, HTML, CSS, Bootstrap, Streamlit, Android Studio

EDUCATION

M.Sc. in Computer Science Anglia Ruskin University, Cambridge, UK	2024
Bachelor of Business Administration – Computer Aided Management Guru Gobind Singh Indraprastha University, Delhi	2020
Advanced Diploma in Animation Arena Animation, Delhi	2021

CERTIFICATIONS

- Cisco Certified Python Developer